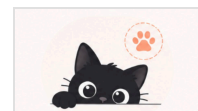


# Project Report

MishiPass — a dynamic QR passport & recovery system for cats



|                                             |                                                            |
|---------------------------------------------|------------------------------------------------------------|
| PROJECT NAME<br>MishiPass (Beta 1.5)        | REFERENCE ID<br>MP-XX-XXXX-XXXX · Version b7f8b204         |
| REPOSITORY<br>github.com/Raven-V1/mishipass | PRODUCTION URL<br>mishipass.carlosvelazquez354.workers.dev |

## 1. Executive summary

MishiPass is a privacy-first dynamic QR passport and recovery system for cats. One permanent QR code adapts to what the cat needs — Active Profile, Missing Alert, Vet Visit, or For Adoption — while keeping private owner and medical data separate from public pages. Built as a Cloudflare Workers TypeScript application with D1 and R2, it delivers a focused product with clear public/private data boundaries, a dynamic QR workflow, and multilingual support in English, Spanish, and Kazakh.

## 2. Project overview

### 2a. Why we're building it

In Mexico, microchipping cats is uncommon, yet there is a strong cat culture and a real need to keep track of them. When a cat goes missing, owners fall back on WhatsApp and Facebook groups to spread the word and coordinate a search. Veterinary records are just as fragile — they are frequently lost or never kept up to date, so critical history disappears exactly when it matters.

MishiPass meets people where they already are: a single printed QR code on the collar, scannable by any phone, backed by a portal that uses existing technology instead of asking owners to adopt a chip reader. That one code becomes the durable record owners lack today — for everyday identification, for a lost-cat alert, for a vet visit, or for a potential adoption. It is **not a substitute for official documentation**; it is a tracker of information that helps owners keep the records needed to fulfill the requirements for official documents when the time comes.

## 2b. How it relates to the theme

MishiPass is built entirely around cats and the people who care for them. It answers the theme with a real-world product that uses Cloudflare's edge platform to serve four distinct experiences from a single static URL — combining Workers, D1, and R2 into one cohesive, privacy-first application grounded in how cat owners actually behave today.

## 2c. Target audience

- Cat owners — especially where chipping is uncommon — who want one collar tag that adapts to their cat's situation
- Finders who scan a found cat's QR and need mode-appropriate information
- Veterinarians documenting visits during temporary Vet Visit sessions
- Cat-owner communities that today rely on WhatsApp and Facebook groups to recover missing cats

# 3. Key features

|                                                                                                                           |                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Dynamic QR routing</b><br>One static URL, four distinct public experiences (Active, Missing, Vet Visit, For Adoption). | <b>Active Profile</b><br>Public-safe cat details with owner-controlled contact.                            |
| <b>Missing Alert</b><br>City/area, reward visibility, sighting reports, WhatsApp card, and Recovery Board.                | <b>Vet Visit</b><br>Temporary, mode-gated form for documentation-only records with 24-hour session.        |
| <b>For Adoption</b><br>Public adoption profile with a transfer-request flow.                                              | <b>Digital Cartilla</b><br>Private owner-only vet visits, vaccines, sticker photos, and Medication Record. |
| <b>Recovery Board</b><br>Public opt-in missing-cat listings with city and alert-age filters.                              | <b>Cat photo gallery</b><br>Multiple photos per cat, owner-only access, selectable profile photo.          |
| <b>Google login</b><br>Logto OIDC with Authorization Code + PKCE, server-side only.                                       | <b>Privacy-first</b><br>No internal IDs, no raw R2 keys, no owner identity on public pages.                |
| <b>Multilingual</b>                                                                                                       |                                                                                                            |

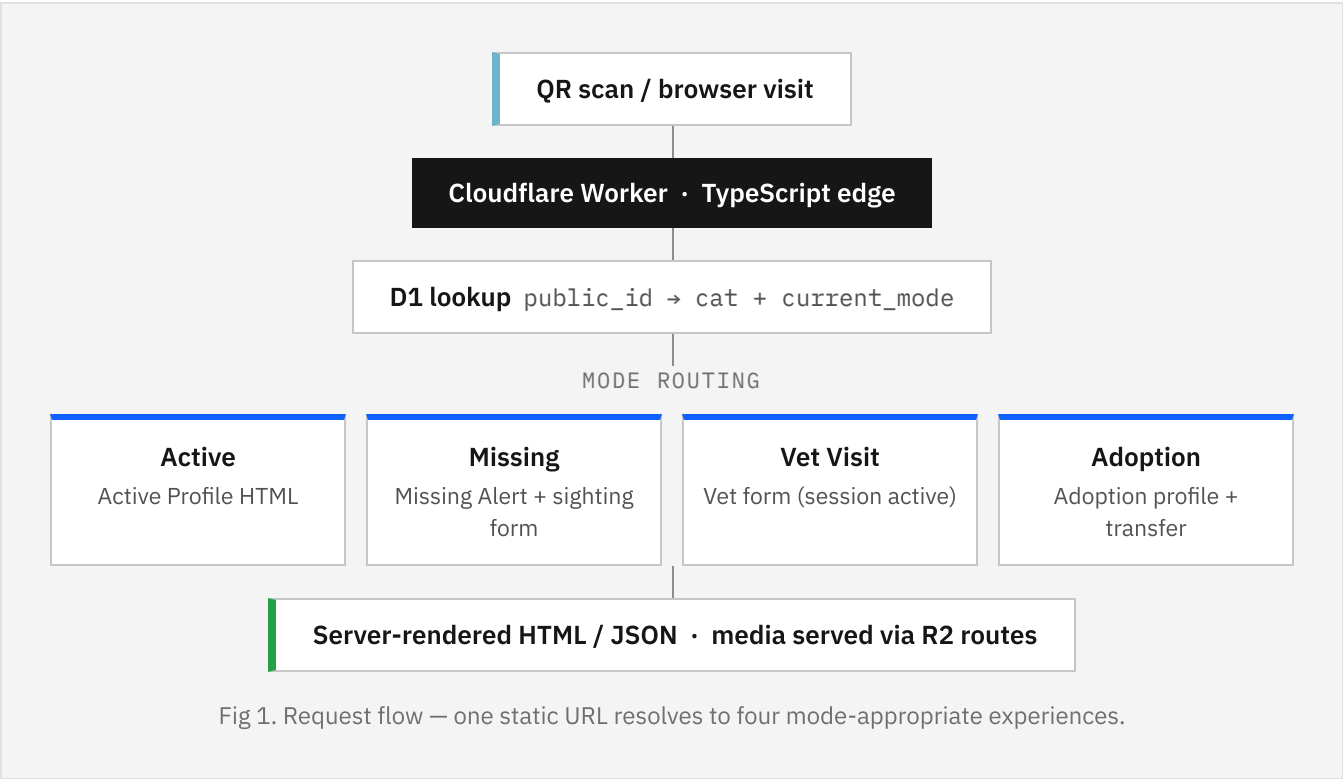
English, Spanish, and Kazakh for both owner and guest interfaces.

## 4. Technology stack

| Layer                 | Technology                                      | Purpose                                           |
|-----------------------|-------------------------------------------------|---------------------------------------------------|
| Runtime               | <a href="#">Cloudflare Workers</a>              | TypeScript edge compute, HTML rendering, and API. |
| Database              | <a href="#">Cloudflare D1</a>                   | Structured data (SQLite-compatible).              |
| File storage          | <a href="#">Cloudflare R2</a>                   | Cat photos, sighting photos, vaccine stickers.    |
| Auth                  | <a href="#">PBKDF2-SHA256 + session cookies</a> | Owner authentication.                             |
| ID format             | <a href="#">Crockford Base32</a>                | ~40-bit public cat IDs.                           |
| CI                    | <a href="#">GitHub Actions</a>                  | Typecheck, tests, and dependency audit.           |
| Dependency monitoring | <a href="#">Dependabot</a>                      | Weekly npm / pip checks.                          |
| Breed assist          | <a href="#">TheCatAPI (optional)</a>            | Reference images for the breed-selection UI.      |

## 5. Technical architecture

The QR encodes one static URL. Every scan reaches the Cloudflare Worker, which resolves the cat's public ID to its current mode in D1 and renders the mode-appropriate server-rendered HTML. The printed QR never changes; only the mode behind it does.



## Data boundaries

| Surface                 | Publicly accessible                                      | Always protected                                    |
|-------------------------|----------------------------------------------------------|-----------------------------------------------------|
| Public /c/:id (active)  | Cat name, country, photo, contact preference             | Owner identity, cartilla, medications, internal IDs |
| Public /c/:id (missing) | Cat name, area, city, reward (if visible), sighting form | Owner identity, cartilla, medications, internal IDs |
| Sighting form           | Submit text + optional photo                             | Reporter IP (HMAC-hashed only)                      |
| Owner dashboard         | — requires an authenticated session —                    | Full profile, cartilla, sightings, gallery          |
| Recovery Board          | Missing alerts (public-safe fields)                      | Owner contact (relay only), medical data            |

## Session & authentication

- PBKDF2-SHA256 password hashing via the Web Crypto API; opaque session token in an HttpOnly cookie, only its SHA-256 hash stored in D1.
- Session expiry enforced server-side; owner-scoped queries prevent cross-owner data access.

- Google login is active via Logto OIDC (Authorization Code + PKCE, server-side). Apple login is a visible placeholder only — the Logto Apple connector is not configured.
- Reporter IPs are HMAC-SHA256 hashed before storage; no raw IP and no raw R2 keys are ever persisted or exposed.

## 6. Testing matrix

Automated coverage: 279 Worker unit tests and 43 shared-validation tests passing, clean `tsc --noEmit` across workspaces, and 0 npm-audit findings. Key manual flows verified against local/staging below.

| Feature / flow         | Steps                                                | Expected result                                                            | Actual result         | Pass / fail |
|------------------------|------------------------------------------------------|----------------------------------------------------------------------------|-----------------------|-------------|
| QR → Active Profile    | Register cat, open /c/:id                            | Active Profile renders with name, country badge, photo, contact preference | Renders as expected   | Pass        |
| Mode switch → Missing  | Switch cat to Missing, re-open same URL              | Missing Alert view with sighting form (different experience, same URL)     | Renders as expected   | Pass        |
| Sighting photo upload  | Submit sighting with image                           | MIME, size, and magic-byte validation enforced; photo stored in R2         | Validation enforced   | Pass        |
| Vet Visit round-trip   | Start Vet Visit, open URL, Save & Finish             | Vet form shows; on finish, cat auto-returns to Active Profile              | Auto-return confirmed | Pass        |
| Recovery Board filters | Open /recovery-board, apply city + alert-age filters | Listings filter correctly by city and age                                  | Filters work          | Pass        |
| Privacy settings       | Toggle relay / phone / hidden                        | Public page reflects chosen contact mode                                   | Reflected correctly   | Pass        |
| Owner-scoped access    | Attempt to view another owner's cat                  | Access denied; cross-owner data not returned                               | Access denied         | Pass        |

## 7. Future improvements

With another week — and beyond Beta — the roadmap focuses on closing the record-keeping loop and reducing manual steps:

- Dedicated veterinarian accounts to replace temporary, scanner-submitted Vet Visit sessions.

- Optional modes — Travel, Memorial, and Celebration — building on the existing mode-routing engine.
- Exportable record summaries to help owners assemble the paperwork required for official documents and travel.
- Individual sighting-report detail views and richer owner inbox management.
- WhatsApp Business backend automation beyond the current manual share link.
- Push notifications for new sightings on cats in Missing Alert mode.

## 8. Tools we used

| Tool                       | Role                                                |
|----------------------------|-----------------------------------------------------|
| Kiro                       | AI implementation assistant                         |
| Codex                      | AI design review and visual-polish assistant        |
| Claude / ChatGPT           | AI advisory review assistants                       |
| Cloudflare Workers, D1, R2 | Production runtime, database, and media storage     |
| GitHub Actions             | CI pipeline (typecheck, tests, audit)               |
| Dependabot                 | Automated dependency monitoring (not a contributor) |
| Wrangler                   | Deployment CLI                                      |

## 9. Learnings & takeaways

- Server-rendered HTML from Workers is fast and avoids client-side JS bundle costs for public pages that need an instant QR-scan response.
- D1's SQLite compatibility works well for a structured product but requires careful migration tooling (semicolon splitting in trigger blocks).
- Privacy-first decisions made early — public IDs, no internal-ID exposure, HMAC IP hashing — save significant rework later.
- Mode-gated access (Vet Visit) is a practical Beta alternative to full account systems when scope and risk are documented.
- Multilingual support is far easier to add during the build than to retrofit afterward.

- Designing around existing behavior — QR, WhatsApp, Facebook groups — lowers adoption friction more than introducing new hardware like chips.

## 10. Acknowledgments

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- Registered team: **Belvenar Analytics Development** — a display-only brand for hackathon participation, not a code contributor.
- Project owner & implementation lead: Carlos Velazquez · Design authority: Zhanerke Askerbekova.
- Platform: Cloudflare Workers, D1, and R2.
- Optional breed reference: TheCatAPI (free tier). Dependency monitoring: Dependabot.
- AI-assisted implementation and review: Kiro, Codex, Claude, and ChatGPT.

## Submission checklist

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- |                                     |                                                                                                                       |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> | Video demo (HD, or at least 720p)                                                                                     |
| <input checked="" type="checkbox"/> | README.md — prerequisites, run instructions, configuration                                                            |
| <input checked="" type="checkbox"/> | Project report (this document, in documentation/)                                                                     |
| <input checked="" type="checkbox"/> | Source code in src/                                                                                                   |
| <input checked="" type="checkbox"/> | No unrelated files, executables, auto-generated code, or package folders (e.g. node_modules/, build/, dist/, vendor/) |